

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

November 2014

EC207 Control Systems

Date: 28.11.2014, Friday

Time: 10:00 a.m. To 01:00 p.m.

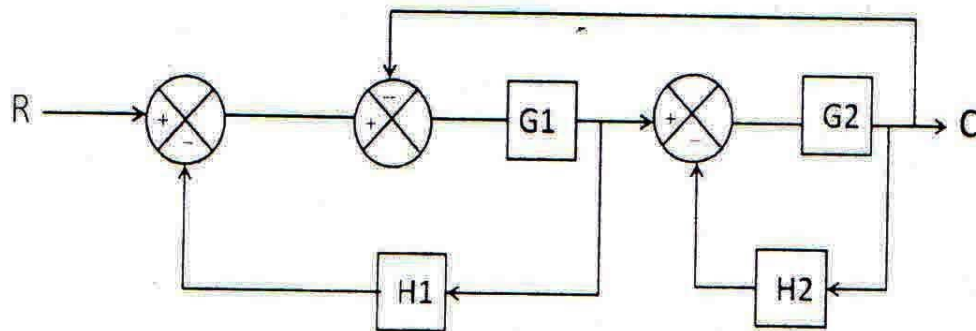
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1(a) Give the differences between block diagram representation and signal flow graph representation of system. [03]
- (b) Discuss the difference between open loop and closed loop control system. [04]
- Q - 2 (a) Using block diagram reduction technique, find the closed loop transfer function of a system given below. [04]



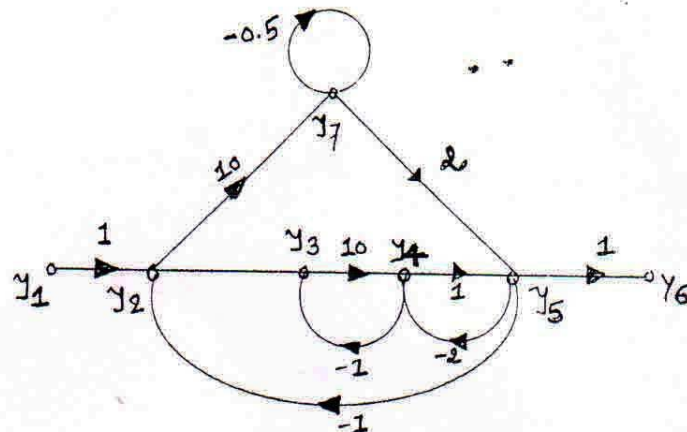
(b) Answer any two questions.

(i) A system described by the state model given below:

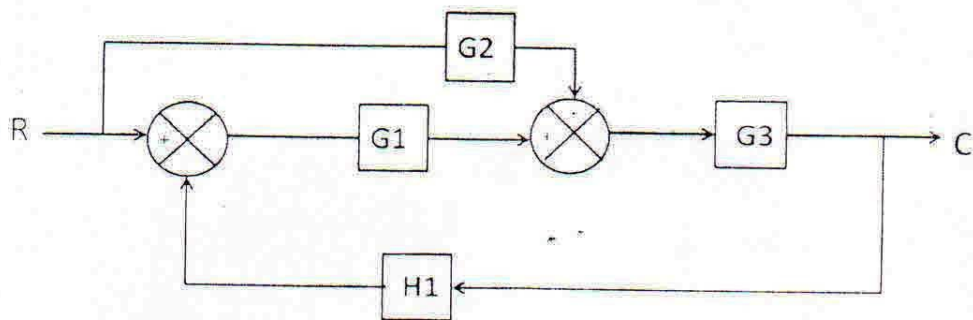
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -12 & -7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$
$$y = \begin{bmatrix} -10 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + u(t)$$

- 1) State Observability.
- 2) State Controllability

- (ii) Obtain transfer function from the signal flow graph shown in following figure (Use Mason's gain formula).



- (iii) For the block diagram shown in figure below, determine the transfer function.



Q - 3 (a) Define Mason's gain formula.

[02]

(b) A Feedback control system has an open-loop transfer function

[07]

$$G(s)H(s) = \frac{K(s+2)}{(s+1)^2}$$

Sketch the Root Locus as K is varied from 0 to ∞ .

OR

(b) Consider a unity feedback control system with an open loop transfer function

[07]

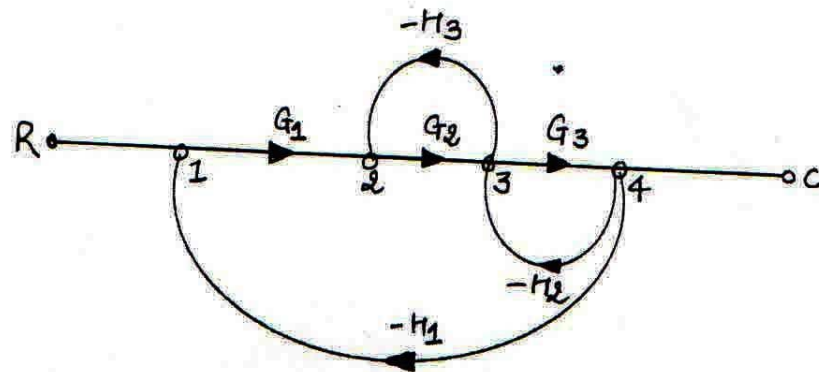
$$G(s)H(s) = \frac{K(s+1)(s+2)}{(s+0.1)(s-1)}$$

Draw the Root Locus for the system with the gain K as variable.

(c) Write short note on Synchro transmitter and receiver.

[05]

- (c) Using Mason's gain formulas obtain transfer function from the signal flow graph shown in figure below. [05]



SECTION - II

- Q - 4 (a) Define the following terms

- 1) Delay time
- 2) Rise time
- 3) Peak time
- 4) Damping
- 5) Peak overshoot
- 6) Settling time
- 7) Steady state error

- Q - 5 (a) Using Routh's criteria determine the stability of a system whose characteristics equation is given by [05]

$$2s^6 + 2s^5 + 3s^4 + 3s^3 + 2s^2 + s + 1 = 0$$

OR

- (a) Find the stability of the system whose characteristics equation is

$$s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50 = 0$$

Also find the roots of this equation.

- (b) How are locations of roots of the characteristic equation related to stability? [03]
- (c) A unit step input is applied to the unity feedback system for which open loop transfer function [06]

$$G(s) = \frac{16}{s(s+8)}$$

Find

- 1) Closed loop transfer function.
- 2) Natural frequency of oscillations.
- 3) Damping Ratio.
- 4) Damped frequency of oscillation.

OR

- (c) A unity feedback system has open loop transfer function given below

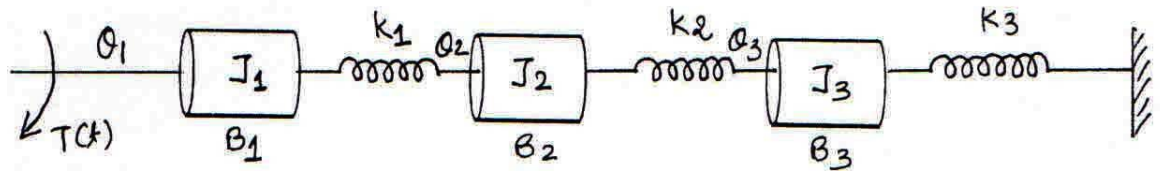
[06]

$$G(s) = \frac{10(s+1)}{s^2(s+2)(s+10)}$$

Determine

- 1) Type of system.
- 2) Error coefficients and
- 3) Steady state error for input as $1 + 4t + \frac{t^2}{2}$.

- Q - 6 (a) Obtain the differential equations for the system shown in figure below and draw the analogue electrical circuit based on torque-voltage analogy. [06]



- (b) Answer any two questions.

[08]

- I. Sketch the asymptotic bode plot for the transfer function given below and also find gain cross over frequency, phase cross over frequency, gain margin and phase margin of the system.

$$G(s)H(s) = \frac{1}{s(s+1)(0.1s+1)}$$

- II. Using Nyquist criterion, examine the closed loop stability of a system whose open loop transfer function is

$$G(s)H(s) = \frac{2500}{s^3(s+300)}$$

- III. Sketch the Nyquist plot for a system with

$$G(s)H(s) = \frac{1}{(s+3)(s+2)}$$

Comment on Stability.
